

TSX Trading Advisor

Technical Reference Manual

Architecture, Configuration, Data Flow, AI Logic and Trading Rules

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Mode: SHORT TERM (1-Week Momentum)

This document is auto-generated and reflects the live codebase. It is intended for the system operator only.

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1. System Overview

TSX Trading Advisor is a zero-cost, single-user autonomous stock trading assistant for Canadian equities on the Toronto Stock Exchange. It continuously monitors a user-defined portfolio, fetches live market data and news, and delivers scheduled AI-generated BUY/SELL/HOLD recommendations via Telegram.

The system runs entirely locally on a Windows machine using Python. There are no cloud services, no subscription fees, and no external AI API keys. All AI reasoning is performed natively by the Antigravity agent session.

Key Characteristics

- Horizon: SHORT TERM (1-week momentum and mean-reversion)
- Data: Live prices, technical indicators, and news fetched every 30 minutes
- Delivery: Telegram Bot messages to the operator's phone
- Control: Operator logs trades via natural language text messages to the bot
- Portfolio: Configurable tickers in config.json
- Scheduling: 30-minute scans (10:00 AM - 3:30 PM ET) + EOD summary (4:05 PM ET)

What the System Does NOT Do

- It does NOT execute trades automatically. All trades must be placed manually.
- It does NOT access broker APIs or brokerage accounts.

(i) IMPORTANT: This tool provides AI-generated trading suggestions only. All investment decisions are the sole responsibility of the operator.

2. Architecture and Data Flow

2.1 High-Level Architecture

The system is split into three distinct layers: a deterministic Data Engine, an AI Reasoning Layer, and a Telegram Delivery Layer. These communicate via local JSONL queue files, creating a clean, fault-tolerant bridge.

```
[Telegram User]
|
v (sends message)
[telegram_daemon.py] <--- Runs 24/7 via start_daemon.bat
|
v (writes to)
[incoming_queue.jsonl]
|
v (monitored by)
[file_watcher.py] <--- Background process, loops continuously
|
v (prints NEW_MESSAGE to stdout, wakes)
[Antigravity Agent] <--- The AI brain
|
+--> runs market_analyzer.generate_dossier()
+--> reasons natively, writes analysis
|
v (writes to)
[outgoing_queue.jsonl]
|
v (polled every 2s by)
[telegram_daemon.py] ---> [Telegram User]
                        +--> log_advice(advice_history.txt)
```

2.2 Scheduled Analysis Flow

Task	Cron Expression	Description
30-min Market Scan	0,30 10-15 ** 1-5	Full dossier + BUY/SELL/HOLD
End-of-Day Summary	5 16 ** 1-5	Portfolio recap + next-day outlook

2.3 Advice History and Memory

Every message sent through `outgoing_queue.jsonl` is automatically logged to `advice_history.txt` by `telegram_daemon.py`. This provides the AI with rolling memory of its own prior advice. The `get_cleaned_advice_history()` function trims the file on each dossier generation to retain: up to the last 5 entries from today plus the final entry from the prior 5 calendar days.

(i) An OS-level file lock (`advice.lock`) prevents corruption from concurrent writes. Stale lock files older than 30 seconds are automatically removed.

3. File Inventory

3.1 Core Python Scripts

File	Role
market_analyzer.py	Core data engine: prices, indicators, dossier, advice history, prompt templates
telegram_daemon.py	Telegram bot relay: receives messages, polls queue, sends responses, logs all advice
file_watcher.py	Queue bridge: loops continuously, watches incoming_queue.jsonl, wakes agent on new
generate_pdf.py	Documentation generator: produces this PDF from source
quiet_watchdog.py	Optional: monitors file_watcher.py process and restarts it if it crashes (checks every 10 min)

3.2 Configuration and Data Files

File	Purpose
config.json	Portfolio tickers, news search terms, operating mode (SHORT/MEDIUM), stop-loss %
trades.csv	Trade ledger: BUY/SELL history, Adjusted Cost Base, running cash balance
advice_history.txt	Rolling log of all AI advice sent to Telegram
.env	Secrets: TELEGRAM_BOT_TOKEN, TELEGRAM_CHAT_ID, TELEGRAM_PIN,
.gitignore	Excludes .env, *.lock, __pycache__, queue files from git
requirements.txt	Python dependencies for pip installation (includes psutil)
start_daemon.bat	Windows launcher: double-click to start telegram_daemon.py
AGENT_RESTORE_INSTRUCTIONS.	Step-by-step AI session restore guide
.agents/AGENTS.md	Persistent agent rules: trading strategy overrides and security protocol

3.3 Runtime Queue Files (transient)

File	Purpose
incoming_queue.jsonl	Telegram messages from user, waiting for AI processing
incoming_processing.jsonl	Atomic rename during processing (prevents double-read)
outgoing_queue.jsonl	AI responses queued for Telegram delivery
outgoing_processing.jsonl	Atomic rename during daemon polling
advice.lock	OS-level mutex protecting advice_history.txt writes

(i) Queue files are ephemeral and not committed to git. Created automatically at runtime.

4. Configuration Reference (config.json)

config.json is the single control plane for the system.

```
{
  "mode": "SHORT",
  "stop_loss_pct": 5.0,
  "portfolio": {
    "CNQ.TO": [
      "CNQ.TO",
      "CL=F",
      "canadian+oil+sector",
      "OPEC+oil+production"
    ],
    "CLS.TO": [
      "CLS.TO",
      "Celestica+electronics",
      "electronics+manufacturing+services",
      "supply+chain+hardware"
    ]
  }
}
```

Fields

Field	Type	Description
mode	string	"SHORT" (1-week momentum) or "MEDIUM" (2-4 week swing). Controls
stop_loss_pct	float	If a position drops this % below ACB, a mandatory SELL directive is injected
portfolio	object	Keys are ticker symbols. Values are arrays of news search topics.

Portfolio News Topics (per-ticker array)

Each entry in the array can be:

- A ticker symbol (e.g. "CNQ.TO") - fetched from Yahoo Finance RSS
- A commodity symbol (e.g. "CL=F") - used as a macro price tracker
- A plain-text search term (e.g. "canadian+oil+sector") - searched on Google News

(i) Use full company names for Google News terms, not bare tickers. "Canadian+Pacific+Kansas+City" returns far more relevant results than "CP".

5. Market Dossier Structure

The dossier is a structured plain-text document generated by `market_analyzer.generate_dossier()` on every scheduled scan or manual analysis request. It contains five sections.

Section 1: Mode and Timestamp

```
=== CURRENT MODE: SHORT TERM (1-WEEK HORIZON) ===
Date: 2026-06-23 14:30:01 ET
```

Section 2: Portfolio and Pricing (per ticker)

Field	Description
Shares Owned	From trades.csv ledger
Avg Purchase Price (ACB)	Adjusted Cost Base computed from trade history
Current Market Price	Live price from yfinance
Previous Close	Prior day closing price
Gap %	Overnight gap: (Open - PrevClose) / PrevClose
Open / High / Low	Intraday OHLC from yfinance
Intraday Trend %	How far current price is above the Low of Day
Unrealised P&L	Dollar and % profit/loss on current position
STOP-LOSS Alert	Injected if price is > stop_loss_pct% below ACB
Vol Ratio (10-day)	Volume divided by 10-day average volume
EMA Crossover	Explicit BULLISH / BEARISH / NEUTRAL label (EMA5 vs EMA9)
EMA (5-day)	5-day Exponential Moving Average
EMA (9-day)	9-day Exponential Moving Average
RSI (7-day)	7-period Relative Strength Index
ATR (5-day)	Average True Range (5-period)
Bollinger Lower/Mid/Upper	20-period Bollinger Bands (2 std deviations)

(i) In MEDIUM mode: RSI (14-day), SMA (50-day), BB (50-period), Ex-Dividend Date are used instead.

Section 3: Macro Commodities

Commodity symbols (e.g. CL=F, GC=F) are shown as a standalone macro context with current price and daily % change vs open.

Section 4: Qualitative News (up to 10 headlines per topic)

- [NEW] = published within the last 4 hours
- [RECENT] = published 4-12 hours ago
- [OLD] = published 12-48 hours ago
- [MACRO], [EARNINGS], [ANALYST], or [NEWS] category label
- 200-character summary from the RSS description field

Section 5: Previous Advice History

A rolling window of past AI analysis. The LLM receives up to the last 5 entries from today plus the final entry from

the preceding 5 calendar days. Weekends are NOT filtered.

6. AI Trading Rules and Logic

Trading rules live in two locations: system prompt instructions in `market_analyzer.get_analysis_prompt()` and persistent override rules in `.agents/AGENTS.md`. AGENTS.md takes precedence.

6.1 System Prompt Rules (SHORT mode)

Rule 1. No Falling Knives

$EMA5 < EMA9$ means a downtrend is active. Do not assume snap-back. Wait for crossover or volume shock before BUY.

Rule 2. Volume Capitulation

$Vol_Ratio > 1.25$ during a drop = panic selling exhausted. Bullish reversal signal.

Rule 3. Market Open Rule

Before 9:30 AM ET, never recommend buying. Wait 30 minutes for HFT volatility to settle.

Rule 4. Buy the Rumour

Anticipate macro events within 24-48h. Override oversold technicals with HOLD/SELL if event removes a market premium.

Rule 5. Cash Constraint

Check Available Free Cash before every BUY. No cash = no BUY.

Rule 6. Tranching

Deploy free cash in tranches (e.g. 25%) instead of all at once.

Rule 7. Stop-Loss Enforcement

$Position > stop_loss_pct\%$ below ACB = MUST SELL, regardless of RSI.

Rule 8. Profit-Taking

Position touches Upper Bollinger Band AND $RSI > 70$ = trim position.

Rule 9. Zero-Share Rule

0 shares + no BUY = AVOID or IGNORE (not HOLD).

Rule 10. Acknowledge Failures

If previous advice was wrong, explicitly say so. Do not repeat a failed thesis.

Rule 11. Macro Override

$RSI < 25$ + top-tier macro catalyst = Speculative Macro Entry allowed (small position).

6.2 AGENTS.md Persistent Override Rules

- Target Sell Price on BUY

Every BUY must include a Target Sell Price (Upper BB or ATR-based) and Stop Loss Price for the broker.

- Low-Volume Scaled Entry

Bullish EMA crossover without high volume: recommend BUY at 25-50% position size, labeled 'Low-Volume Scaled Entry'.

- Momentum Breakout Exception

Bullish EMA crossover + $RSI > 60$ + green sector: Low-Volume Scaled Entry at 25% allocation still allowed.

- Time-of-Day Filter

Never initiate Low-Volume Scaled Entry between 11:00 AM and 2:00 PM ET.

- Intraday Reversal Trigger

Drop >1.25% below open + RSI < 35 + rebounds within two 30-min intervals: speculative Reversal Buy on partial position.

7. Trading Prompt System Instructions

The AI receives one of two prompt templates selected by config.json mode. Both embed the full dossier and all system rules inline.

7.1 SHORT Mode Output Format

- Brief introduction (1 sentence)
- Prioritised action list: e.g. '1. SELL CNQ.TO, 2. BUY ABX.TO' or 'No immediate actions recommended'
- Detailed breakdown ONLY for action stocks (max 3 sentences per stock, must include current price)

7.2 EOD Summary Output Format

- End-of-Day Market Summary (1-2 sentences)
- Portfolio Performance: status of all held assets (ignore 0-share positions)
- Outlook Ahead: 2-3 sentences on tomorrow based on price action, volume, and macro news

(i) The EOD prompt explicitly instructs the AI NOT to recommend new trades.

8. Telegram Security Protocol

Because the Telegram bot relays messages directly to the AI agent which can run commands and edit files, a PIN challenge protocol is enforced to prevent unauthorized remote code execution.

8.1 Triggers Requiring PIN Verification

- Editing any .py source file
- Editing .agents/AGENTS.md or .env
- Running arbitrary system or PowerShell commands
- Any action modifying the codebase or host system

8.2 PIN Challenge Flow

```

[Telegram: code change request]
  |
  v
AI writes PIN challenge to outgoing_queue.jsonl
(includes TELEGRAM_PIN_HINT from .env)
  |
  v
[User replies with 8-digit PIN]
  |
  v
AI validates against TELEGRAM_PIN in .env
Correct: proceed | Wrong: refuse and log

```

8.3 Local User Exception

Requests from the local Antigravity interface (USER_REQUEST tags) are fully trusted. No PIN required.

8.4 Environment Variables

Variable	Description
TELEGRAM_BOT_TOKEN	API token from @BotFather. Authenticates with the Telegram API.
TELEGRAM_CHAT_ID	Numeric chat ID of the operator. All other chat IDs are rejected.
TELEGRAM_PIN	8-digit PIN for remote code change authentication.
TELEGRAM_PIN_HINT	Hint phrase sent to user when PIN challenge is issued.

(i) NEVER commit .env to git. Listed in .gitignore. Back up your bot token and PIN securely.

9. Session Restore Procedure

If the Antigravity agent session is lost (reboot, context expiry), restore in this order:

Step 1: Start the Telegram Daemon (User Action)

Double-click start_daemon.bat, or run:

```
C:\Users\Aamir\miniforge3\python.exe telegram_daemon.py
```

Step 2: Start the File Watcher (Agent Action)

The agent starts file_watcher.py as a background task:

```
C:\Users\Aamir\miniforge3\python.exe file_watcher.py
```

(i) file_watcher.py now runs in a continuous loop and does NOT exit after one message (fixed in v1.0.1).

Step 3: Recreate Cron Schedules (Agent Action)

Task	CronExpression	Prompt Summary
30-min Market Scan	0,30 10-15 * * 1-5	[30-min Scan] Dossier + analysis ->
EOD Summary	5 16 * * 1-5	[EOD Summary] Dossier + EOD summary -> outgoing_queue.jsonl

Step 4: Verify

- Send 'Analyze' via Telegram - agent should respond within 30 seconds
- Send '/status' via Telegram - daemon replies with uptime and queue sizes
- Check advice_history.txt is being updated after each analysis

10. Known Limitations and Operational Notes

10.1 Market Data

- yfinance prices are delayed approximately 15 minutes for TSX equities.
- EMA, RSI, ATR indicators lag by one bar (prior day's close). The live price is current but indicators are historical.
- Intraday High/Low/Open reflect the current trading day only.

10.2 News Quality

- Google News RSS is keyword-based. Use specific company names to minimize irrelevant results.
- Yahoo Finance RSS is limited to the ticker's own headlines.
- News older than 48 hours is automatically filtered.

10.3 Advice History

- LLM context window receives last 5 entries from today + final entry from prior 5 calendar days.
- Weekends are NOT filtered (fixed in v1.0.1). Friday advice is preserved into Monday.
- advice_history.txt retains full history for human audit.

10.4 Operational Constraints

- System requires the Antigravity agent session to be active during market hours.
- If the session is lost, incoming_queue.jsonl will accumulate unprocessed messages until restored.
- quiet_watchdog.py monitors file_watcher.py only - does NOT monitor the daemon or AI session.
- Very long AI responses are automatically chunked at 4,000 characters per Telegram message.

10.5 Trade Logging

- All trades are logged manually via natural language Telegram messages.
- Cash balance is always read from the Cash_Balance column of the LAST row in trades.csv.
- ACB is computed by rolling forward every row in trades.csv from the beginning.
- Do not insert rows in the middle of trades.csv - this will corrupt the cash balance.

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